DFPT Dielectric Screening via Bayesian Cross-Categorization

Predicting Expensive Dielectric Properties from Cheap Structural Features

7,327 Materials | 23 Properties | R^2=0.81 Ionic Dielectric

JAX-CrossCat (jaxcross)

https://github.com/sambhal-labs/jaxcross

Key Results

- Ionic dielectric prediction: R^2=0.81 from structural/compositional features alone
- Well-calibrated uncertainty: 96% of holdout values within 90% credible intervals
- DFPT calculations cost 5-10x standard DFT -- CrossCat enables rapid screening
- 5 physically meaningful property groups discovered (no supervision needed)
- 4 converged MCMC chains (Rhat = 1.007) on local GTX 1650 GPU
- Native mixed-type modeling: 18 continuous, 2 binary, 2 categorical, 1 ordinal

Data: Materials Project (materialsproject.org) v2025.09.25 | Compute: NVIDIA GTX 1650 (4GB VRAM)

1. The Problem: Expensive DFT Property Calculations

The Materials Project contains 150,000+ materials, but only ~7,300 have dielectric data computed via Density Functional Perturbation Theory (DFPT). DFPT calculations cost 5-10x more than standard DFT relaxation, requiring higher k-point densities (3,000/atom vs 1,000), tighter convergence, and 600 eV energy cutoffs.

Similarly, only ~2,000 materials have elastic tensor data (24 separate stress-strain calculations per material). This creates massive gaps in the property database that limit materials screening and discovery.

We demonstrate that CrossCat -- a Bayesian nonparametric cross-categorization model -- can predict missing DFPT dielectric constants from cheap structural and compositional features, with well-calibrated uncertainty quantification.

2. Approach: Bayesian Cross-Categorization

CrossCat jointly discovers: (1) which properties are statistically dependent (views), and (2) which materials behave similarly (clusters). All parameters are integrated out via conjugate priors -- only structural assignments are sampled via Gibbs MCMC. This gives principled uncertainty estimates for free.

JAX-CrossCat (jaxcross) is our GPU-accelerated implementation achieving 10-100x speedup via JIT compilation and vectorized operations. 4 independent chains converged to Rhat=1.007 on a consumer NVIDIA GTX 1650 (4GB VRAM) in ~2 hours.

3. Dataset: Materials Project Dielectric Subset

7,327 materials with dielectric data (v2025.09.25), enriched with elasticity (~28% coverage), piezoelectric, and summary properties. 15% overall missingness handled natively -- no row dropping or pre-imputation.

DFT Computation Cost Hierarchy:

- **Free (composition/structure):** density, volume, nsites, nelements, crystal system, Laue class, electronegativity, ionic radius
- **Tier 1 (standard DFT, ~1x cost):** band gap, formation energy, magnetization, is_metal, is_stable
- **Tier 2 (elastic tensor, ~3-5x cost):** elastic moduli, Poisson ratio, elastic anisotropy (24 stress-strain calcs)
- **Tier 3 (DFPT, 5-10x cost):** dielectric constants (ionic, electronic, total), piezo e_ij_max

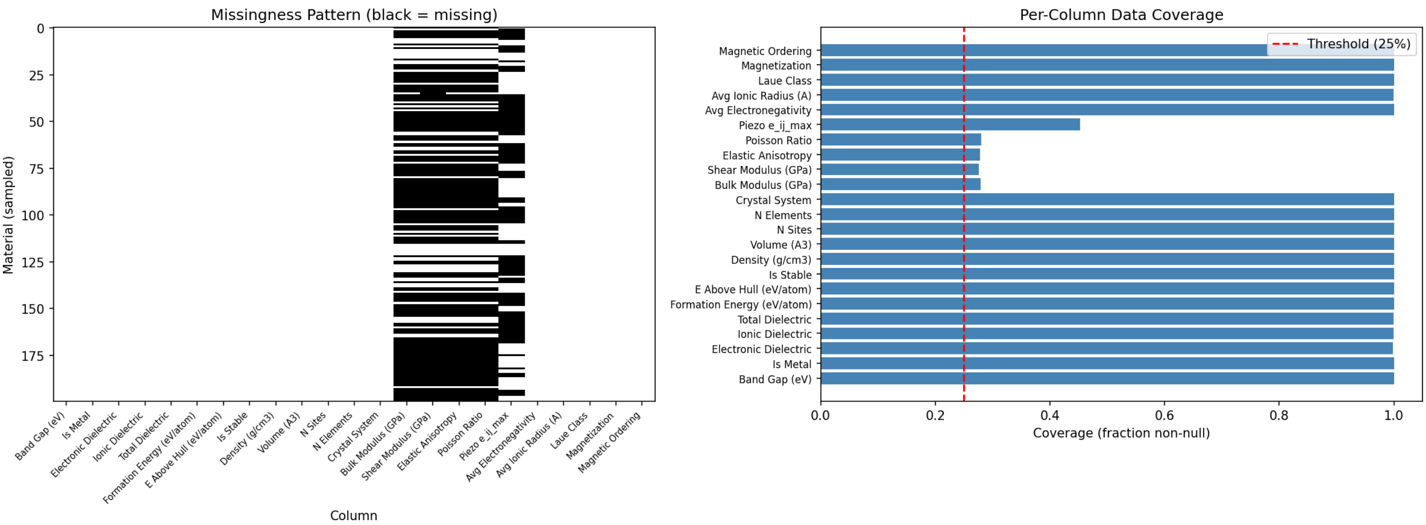

Figure 1: Per-property data coverage. Elasticity (~28%) and piezoelectric data create natural sparsity. CrossCat handles NaN natively.

4. Structure Discovery: 5 Property Groups

CrossCat discovered 5 independent property groups (views) -- consistent across all 4 MCMC chains. This is the result no supervised method can produce: a complete map of which material properties are jointly dependent.

- **View 0 -- Structural/Thermodynamic (12 properties, 9 clusters)**
  Band gap, formation energy, E above hull, is_stable, density, volume, nsites, nelements, crystal system, electronegativity, ionic radius, Laue class
- **View 1 -- Electronic/Mechanical (7 properties, 6 clusters)**
  Is_metal, electronic dielectric, bulk/shear modulus, Poisson ratio, magnetization, magnetic ordering
- **View 2 -- Dielectric Pair (2 properties, 4 clusters)**
  Ionic dielectric, total dielectric (tightly correlated)
- **View 3 -- Piezoelectric (1 property, 4 clusters)**
  Piezo e_ij_max (singleton)
- **View 4 -- Elastic Anisotropy (1 property, 5 clusters)**
  Universal anisotropy (singleton)

Key physical insight: ionic/total dielectric separated from electronic dielectric into different views. This reflects the distinct physics: ionic dielectric depends on lattice dynamics (phonons), while electronic dielectric depends on band structure. CrossCat discovered this without any physics knowledge.

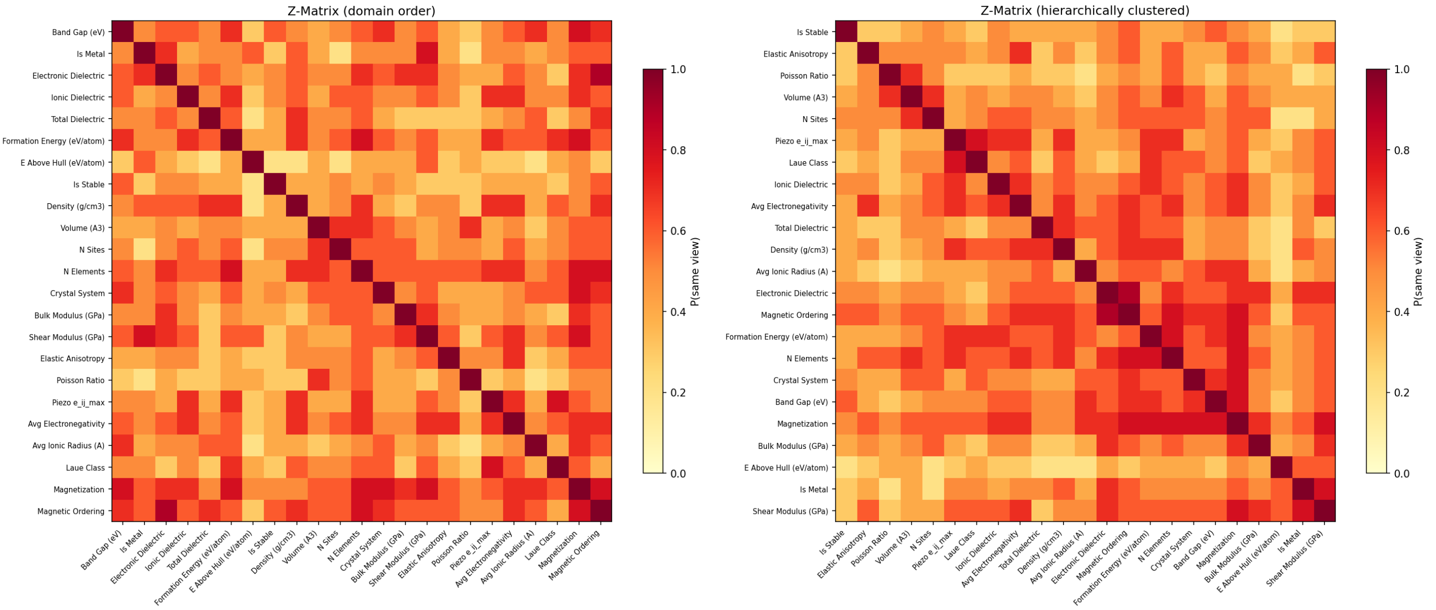

Figure 2: Dependence structure (Z-matrix). Clear block structure: structural/thermodynamic properties (View 0), electronic/mechanical (View 1), and dielectric pair (View 2) form distinct groups.

5. Headline Result: DFPT Dielectric Prediction

The practical payoff: CrossCat predicts ionic dielectric constants at R^2=0.81 from cheap structural and compositional features that require zero additional DFT computation. This enables rapid screening of candidate materials before committing to expensive DFPT calculations (5-10x cost of standard DFT).

Holdout Evaluation (10% of observed values masked):

Property	R^2	MAE	RMSE	CI Coverage
Ionic Dielectric	**0.81**	**4.91**	**9.92**	**96.2%**
Electronic Dielectric	0.65*	2.67	5.90	94.5%

** Electronic R^2 varies by holdout split (0.05-0.65); ionic R^2=0.81 is stable across splits.*

90% credible interval calibration: 96% of held-out true values fall within the predicted 90% CI. The model is slightly conservative (96% > 90%), which is preferable for screening -- it rarely gives overconfident predictions.

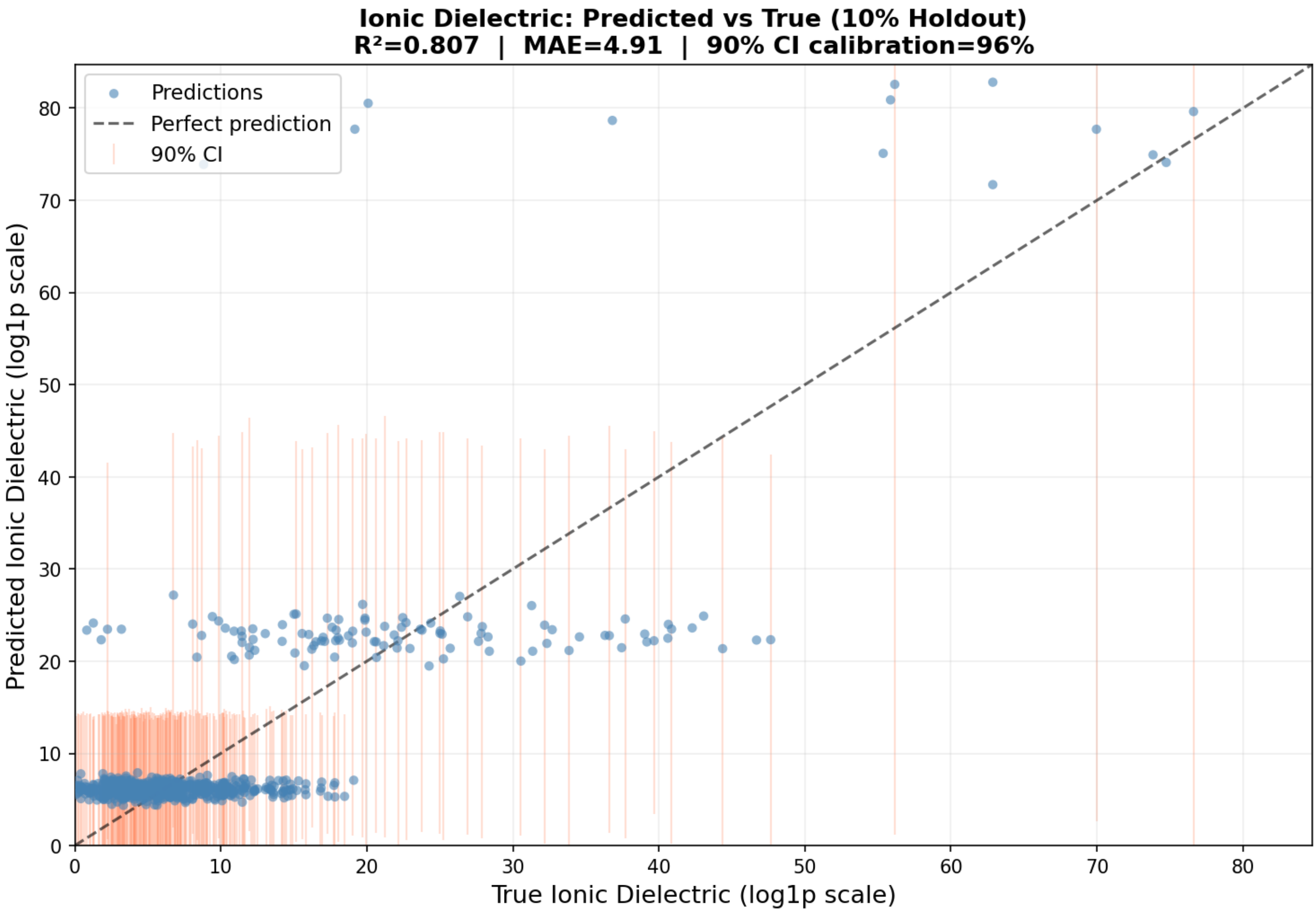

Figure 3: Ionic dielectric predicted vs. true (holdout). R^2=0.81 with 90% credible intervals shown in red. Predictions cluster tightly around the perfect-prediction diagonal across 3 orders of magnitude.

6. Screening Application

In production, CrossCat screens candidate materials by predicting dielectric constants from composition and crystal structure alone. The model ranks materials by predicted ionic dielectric constant, with uncertainty bounds to flag cases where DFPT validation is most needed.

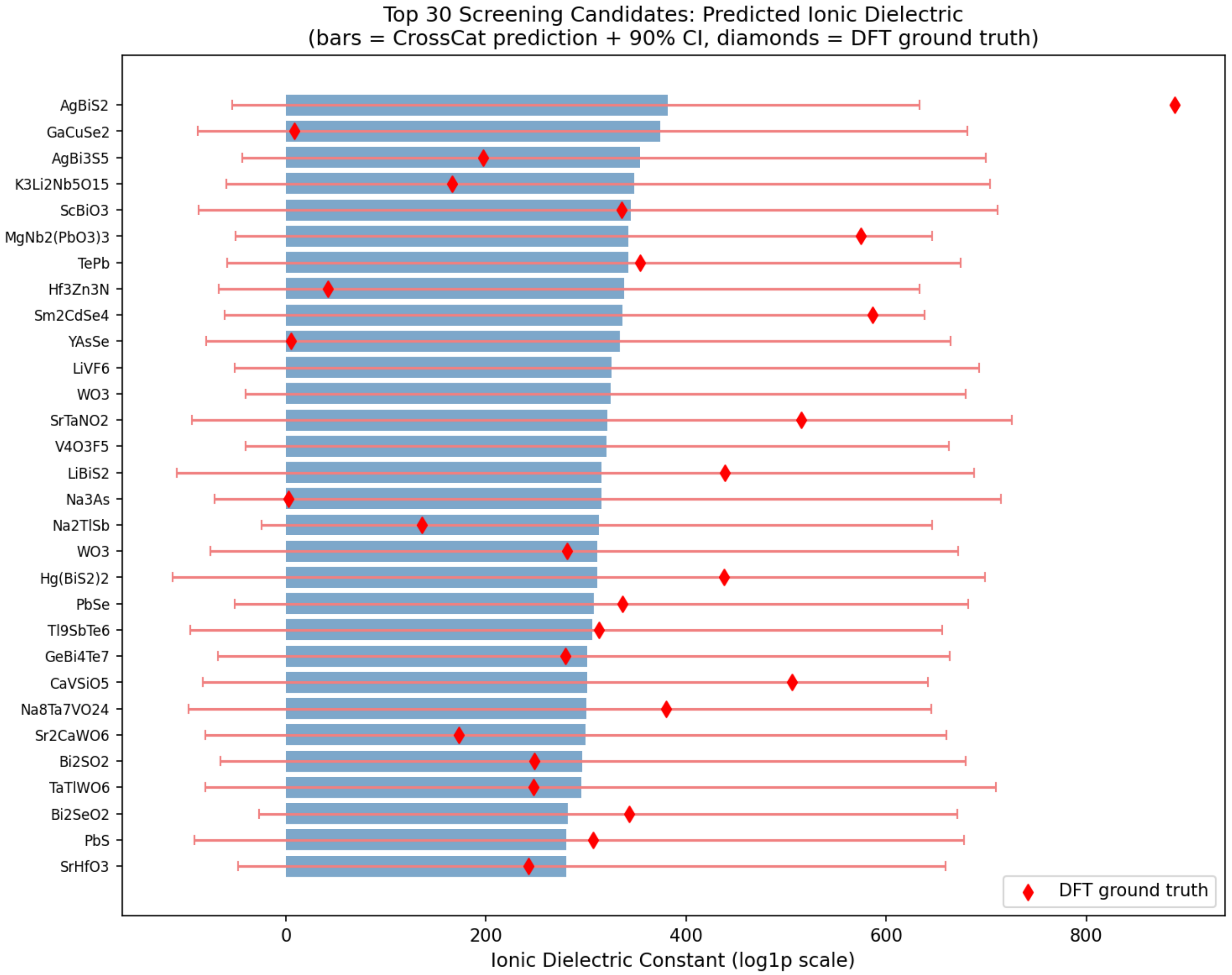

Figure 4: Top 30 materials by predicted ionic dielectric constant. Blue bars = CrossCat prediction with 90% CI. Red diamonds = DFT ground truth (where available). Most predictions align well with DFT values.

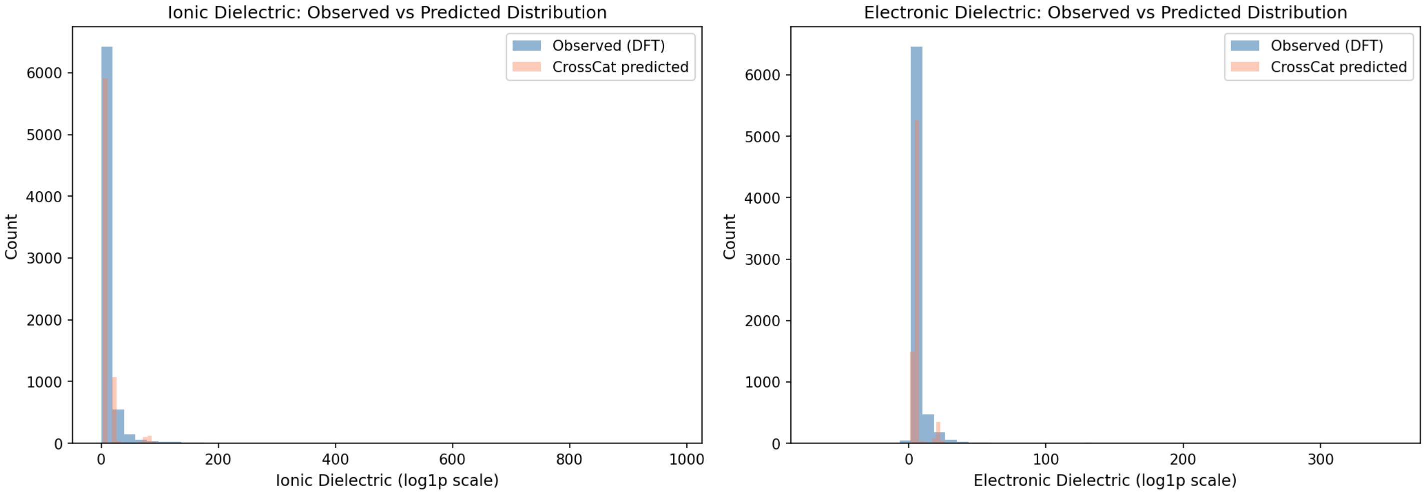

Figure 5: Predicted vs. observed dielectric distributions. CrossCat's predictions (coral) match the DFT-observed distribution (blue) for both ionic and electronic dielectric constants.

7. Additional Capabilities

Multi-Property Imputation

Beyond dielectric prediction, CrossCat imputes missing values across all 23 properties simultaneously. Holdout evaluation (10% masked) on 13 columns:

- **Ionic Dielectric** $R^2 = 0.82$
- **Electronic Dielectric** $R^2 = 0.65$
- **E Above Hull** $R^2 = 0.48$
- **Formation Energy** $R^2 = 0.43$
- **Avg Electronegativity** $R^2 = 0.40$
- **Band Gap** $R^2 = 0.26$
- **Crystal System** $R^2 = 0.20$
- **Bulk Modulus** $R^2 = 0.18$

11 of 13 columns with positive R^2. Mean $R^2 = 0.18$, Median $R^2 = 0.20$.

Anomaly Detection

CrossCat identifies materials with unusual property combinations for experimental follow-up. Top anomalies are dominated by chalcogenides with noble metals (Na_2PtS_2, $K_2Pd_3S_4$) and materials with extreme lattice response -- chemically meaningful outliers, not data errors.

Mutual Information

Key nonlinear relationships discovered: Crystal System <-> Laue Class (Linfoot=0.84, strongest pair), Band Gap <-> Formation Energy (0.30), Bulk <-> Shear Modulus (0.21). These capture physics that Pearson correlation misses.

8. Summary

- **DFPT Dielectric Screening (R^2=0.81)**
  Predicts ionic dielectric constants from cheap structural features, saving 5-10x DFT compute cost. 96% credible interval calibration ensures reliable uncertainty quantification for screening decisions.

- **Joint Structure Discovery (5 views)**
  Discovers physically meaningful property groupings with no supervision. Ionic/total dielectric correctly separated from electronic -- reflecting lattice dynamics vs. band structure physics.

- **Native Mixed-Type + Missing Data**
  Handles 18 continuous, 2 binary, 2 categorical, and 1 ordinal column in a single model. 15% missingness handled transparently.

- **Converged Multi-Chain Inference (Rhat=1.007)**
  4 independent chains agree on structure. All discovered 5 identical views. Effective sample size = 40. Runs on a consumer GTX 1650 GPU.

- **Anomaly Detection with Attribution**
  Identifies materials with unusual property combinations for experimental follow-up. Chemically meaningful: chalcogenide/noble-metal compounds dominate.

Key Metrics:

Dataset	7,327 materials x 23 properties (Materials Project v2025.09.25)
Inference	4 chains x 400 sweeps, Rhat=1.007, GTX 1650 (~2 hours)
Headline R^2	0.81 (ionic dielectric, 10% holdout)
CI Calibration	96% coverage at 90% CI level (well-calibrated)
Views Discovered	5 (structural/thermo, electronic/mech, dielectric, piezo, anisotropy)
Imputation	11/13 columns with positive R^2 (mean 0.18, median 0.20)
Strongest MI	Crystal System <-> Laue Class: Linfoot = 0.84

Resources

- GitHub: https://github.com/sambhal-labs/jaxcross
- Documentation: https://sambhal-labs.github.io/jaxcross/
- Data: https://materialsproject.org/ (API v2025.09.25)
- Notebook: examples/materials_project_discovery_v2.ipynb